

Task 1: Build & Evaluate a Linear Regression Model

California Housing Dataset — House Price Predictor

Submitted by	Krish Choudhary
Internship Role	AI & ML Intern
Date	June 2025
Tools Used	Python, pandas, scikit-learn, matplotlib, seaborn

1. Objective

The goal of this task was to go through the full machine learning workflow — from loading and exploring data, to training a model and evaluating its performance. I used the California Housing dataset and built a Linear Regression model to predict median house values. This task helped me understand how each step of the ML pipeline fits together.

Item	Details
Dataset	California Housing (sklearn built-in)
Samples	20,640 rows, 8 features + 1 target
Target	MedHouseVal — Median House Value (in \$100k units)
Train/Test	80% training, 20% testing (random_state=42)

2. Exploratory Data Analysis

Before building the model I spent time understanding the data. The dataset has no missing values, which made preprocessing straightforward. I checked distributions, looked at correlations between features, and identified which features are most related to house prices.

2.1 Target Variable Distribution

The target column MedHouseVal is right-skewed and capped at 5.0 (\$500k). This clipping is visible in the histogram as a spike at the right end. Most houses fall in the \$100k–\$400k range.

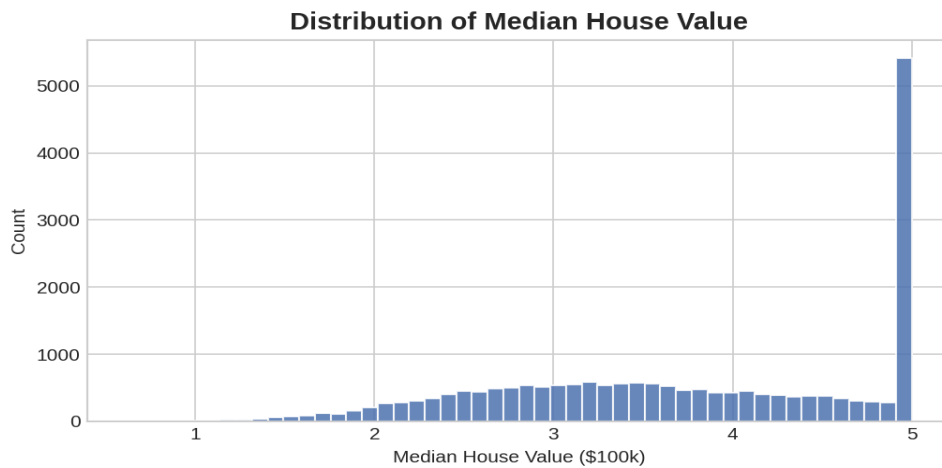


Figure 1: Distribution of Median House Value — capped at 5.0 (\$500k)

2.2 Correlation Heatmap

From the heatmap, MedInc (Median Income) clearly has the highest positive correlation with the target (~0.69). AveBedrms and Latitude show a slight negative correlation. Most other features have weak correlation with house value on their own.

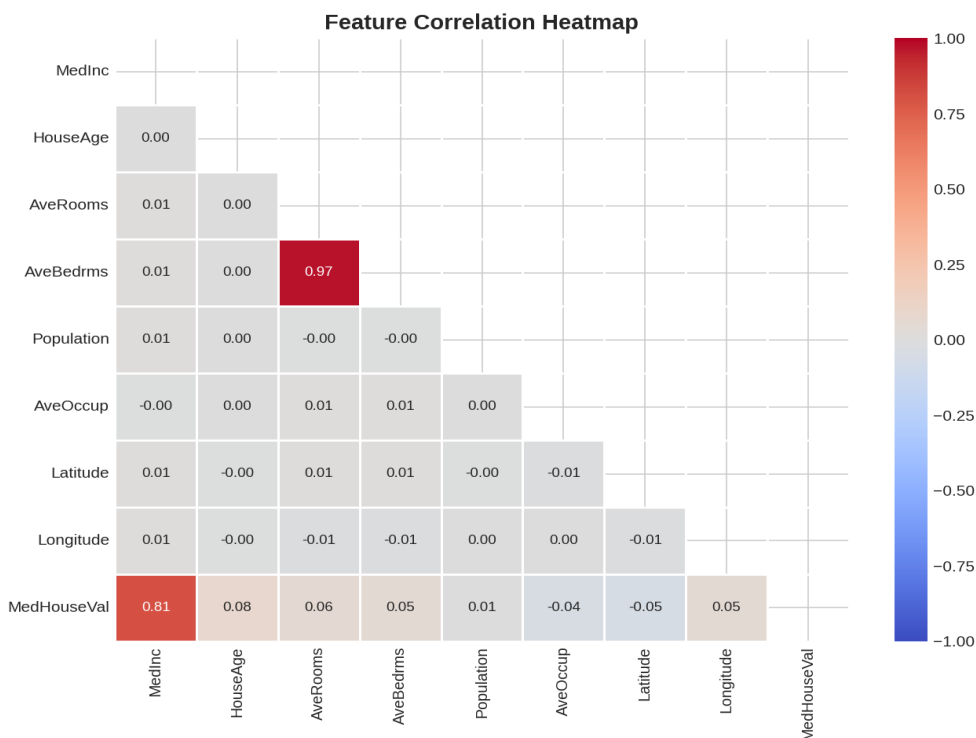


Figure 2: Correlation Heatmap — MedInc stands out as the strongest predictor

2.3 Income vs House Value

Plotting MedInc against the target confirms a clear positive trend — higher income neighbourhoods have more expensive houses. This makes practical sense and suggests the feature will be important in the model.

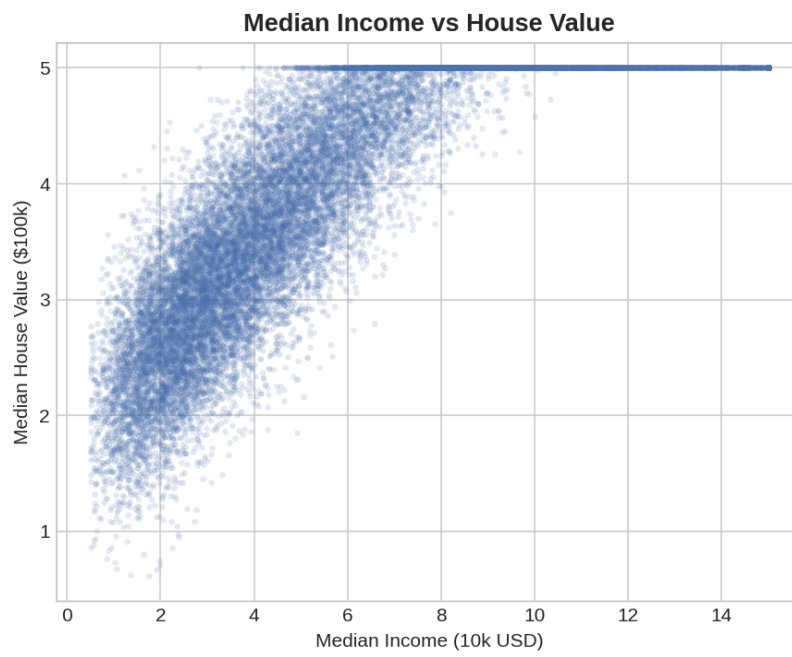


Figure 3: Median Income vs House Value — strong positive relationship

3. Model Training

I used sklearn's LinearRegression with all 8 features. No scaling was applied since linear regression doesn't strictly require it (though scaling would help with interpretability of coefficients). The model was trained on 16,512 samples.

Setting	Value
Algorithm	LinearRegression (Ordinary Least Squares)
Features	MedInc, HouseAge, AveRooms, AveBedrms, Population, AveOccup, Latitude, Longitude
Fit Intercept	True
Test Split	20% — 4,128 test samples

4. Results & Evaluation

4.1 Metrics

Metric	Score	What it means
MAE (Mean Absolute Error)	0.4848	Avg prediction off by ~\$48480
RMSE (Root Mean Sq. Error)	0.5862	Penalises large errors more than MAE
R ² Score	0.6732	Model explains 67.3% of price variance

An R² of 0.6732 is a decent result for a basic linear model with no feature engineering. The MAE tells us predictions are typically off by around \$48,000 — not great for high-value properties but reasonable as a starting point.

4.2 Actual vs Predicted

In the plot below, each dot represents one house in the test set. Points closer to the red diagonal line are more accurate predictions. The model works reasonably well for mid-range houses but underpredicts expensive ones (those capped at 5.0).

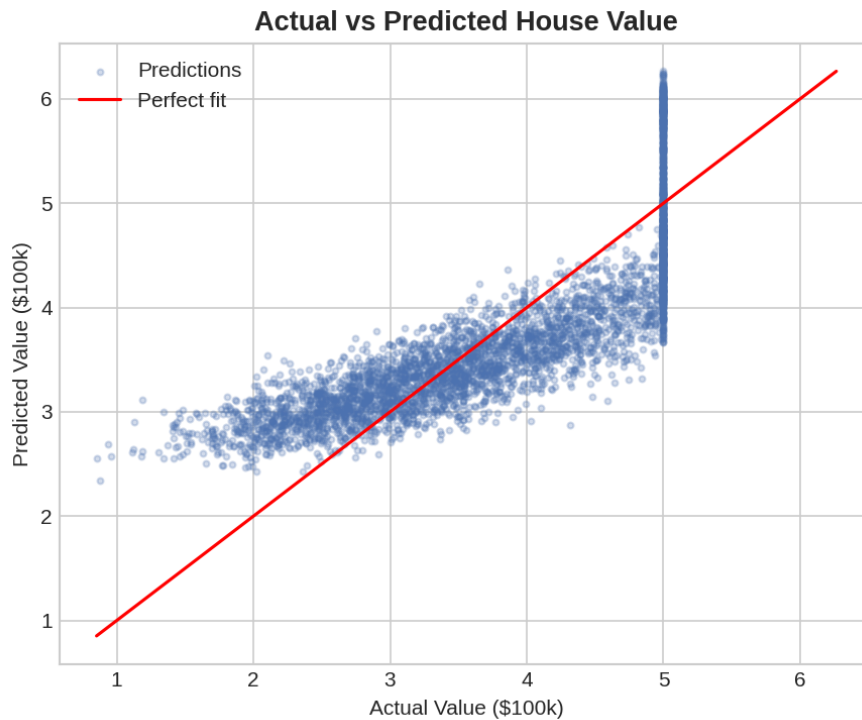


Figure 4: Actual vs Predicted — red line is perfect prediction

4.3 Residual Analysis

The residual plot shows errors scattered around zero for most predictions, which is what we want. However there's a wider spread at higher predicted values — this is called heteroscedasticity and means the model is less reliable for expensive houses. The residual histogram looks roughly bell-shaped, which is a good sign.

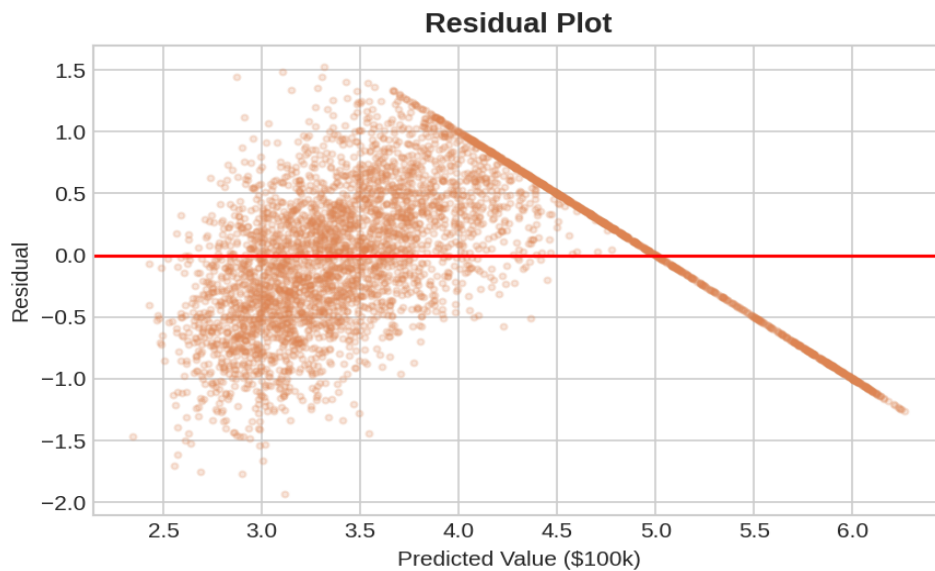


Figure 5: Residuals — mostly centered at 0 but spreads at higher values

4.4 Feature Coefficients

The bar chart below shows the coefficient assigned to each feature. MedInc has the largest positive coefficient, confirming income is the biggest driver. Latitude and Longitude have negative coefficients, reflecting that more southern/coastal locations in California tend to be pricier (larger negative longitude

= further west).

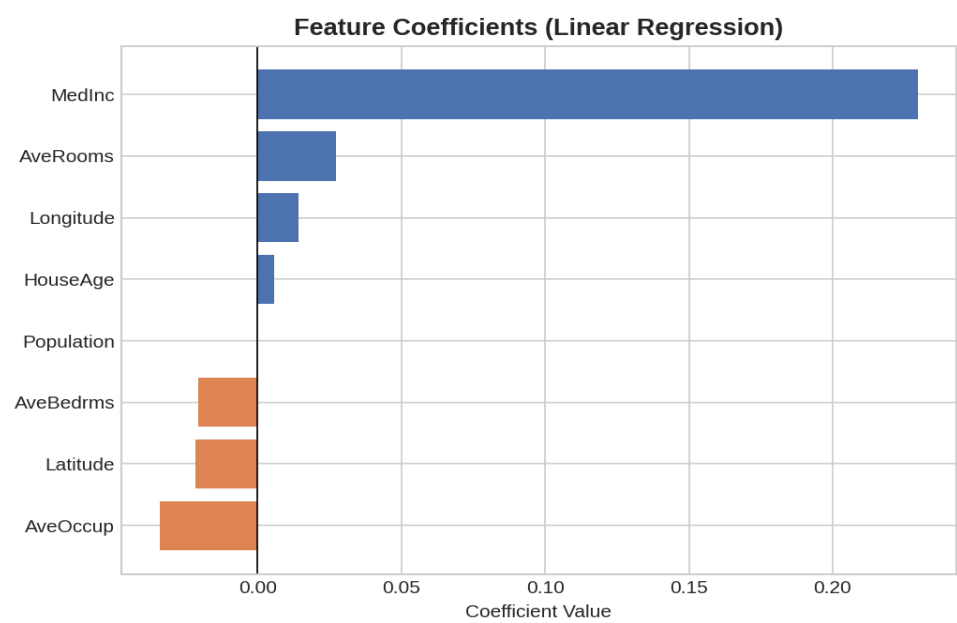


Figure 6: Feature Coefficients from Linear Regression

5. Possible Improvements

The linear regression baseline is decent but there's definitely room to improve. Here are some things I would try next:

- Apply log transformation on skewed features like Population and AveOccup before training
- Add interaction terms — for example MedInc multiplied by HouseAge might capture something useful
- Try Ridge or Lasso regression which add regularization to reduce overfitting
- Use polynomial features (degree 2) to model non-linear relationships
- Try tree-based models like Random Forest or XGBoost which usually get $R^2 > 0.85$ on this dataset
- Cross-validation (k-fold) instead of a single train/test split for more reliable evaluation

6. Conclusion

This task gave me hands-on experience with the full ML workflow. I loaded and explored the California Housing dataset, identified MedInc as the most important predictor, and trained a Linear Regression model that achieved an R^2 of 0.6732 with MAE of 0.4848. The model performs well for average-priced homes but struggles with high-value properties due to the capped target values. Going forward, feature engineering and ensemble models would likely push the R^2 above 0.80.